

Assignment 1

Please answer the following. Answer on a physical copy, whole yellow paper or bond paper (can be more than one paper), any black or blue pen, as long as it is **readable**. Label your answers and section them in terms of struct, array, and linked list answers.

For coding questions, as much as possible, try not to use compilers and try to compile it by yourself.

STRUCTURES

Reference the following structures, answer the following prompts with two variations: firstly, access the data without a pointer and secondly access with a pointer.

<pre>#define MAX 30; #define char String[50]; typedef struct{ String bookName; String authorName; String publishingHouse; int publishingYear; int totalPageNum; }Book;</pre>	<pre>typedef struct{ Book books[20]; int bookCount; }Shelf; typedef struct{ Shelf shelves[3]; int totalBookCount; }BookShelf;</pre>	<pre>BookShelf BS; BookShelf *BointR;</pre>
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Example.

Given the following:

```
Studtype stud;
studPtr ptr = &stud;
```

1. Display the year level of the student.

```
w/o pointer: printf("%d",stud.YrLvl);
w/ pointer: printf("%d",ptr->YrLvl);
```

1. Display the bookName of the 3rd book of the top shelf
2. Update the publishing year of the last book of the bottom shelf
3. Display how many books there are on the top and middle shelf
4. Access the 7th letter of the authorName of the 6th book of the bottom shelf
5. Update the 2nd book on the middle shelf because a new copy came in with 400 total pages
6. Display the 1st letter of the publishingHouse of the last book on the top shelf
7. The 7th book on the bottom shelf is wrong, its author should be "J.R.R. Tolkein"
8. The librarian needs the total book count of the entire bookshelf but its variable is currently empty, what will you do?
9. Insert at the last position, a book Moby into the middle shelf
10. Interchange the first and last book of the top shelf

ARRAYS (using the same structure)

You must initialize the variables before calling the function.

1. Given a BookShelf BS, the function will return 0 if all the books in the top shelf are not unique. If they are, then the function will return 1.

a. Function Header (2pts)

b. Function Call (4pts)

c. Code of the Function (10pts)

d. **Simulation** (4pts) *(Execution Stack is advised but if you can simulate it in another way of drawing, its okay)*
2. Given a BookShelf * Bointer and a string variable of w, the function will delete all occurrences of the specified publishingHouse.

a. Function Header (2pts)

b. Function Call (4pts)

c. Code of the Function (10pts)

d. **Simulation** (4pts)

LINKED LIST

<pre>#define MAX 30; #define char String[50]; typedef struct{ String bookName; String authorName; String publishingHouse; int publishingYear; int totalPageNum; }Books;</pre>	<pre>typedef struct node{ Books book; Struct node* lnik; }Shelf; typedef struct{ Shelf *BookNook; int totalBookCount; }BookShelf;</pre>
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1. Given that BookShelf BSN and an array of books with a size of 5, the function will insert the book in its proper position.

a. Function Header (2pts)

b. Function Call (4pts)

c. Code of the Function (10pts)

d. **Simulation** (4pts)